INCH-POUND
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DETAIL SPECIFICATION

SPEEDOMETERS, TACHOMETERS, FLEXIBLE SHAFT ASSEMBLIES, FLEXIBLE SHAFT ASSEMBLY CORES AND ADAPTERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers mechanical speedometers and tachometers, flexible shaft assemblies, flexible shaft assembly cores and adapters.

1.2 Classification. The speedometers, tachometers, shaft assemblies, cores and adapters are of the groups and classes as specified below (see 6.2).

1.2.1 Groups. The groups of speedometers, tachometers, flexible shaft assemblies, cores and adapters are as follows:

Group A - Speedometer, 0-60 miles per hour (mph), fixed mounting
Group B - Tachometer, fixed mounting
Group F - Flexible shaft assemblies, flexible shaft assembly cores, and adapters

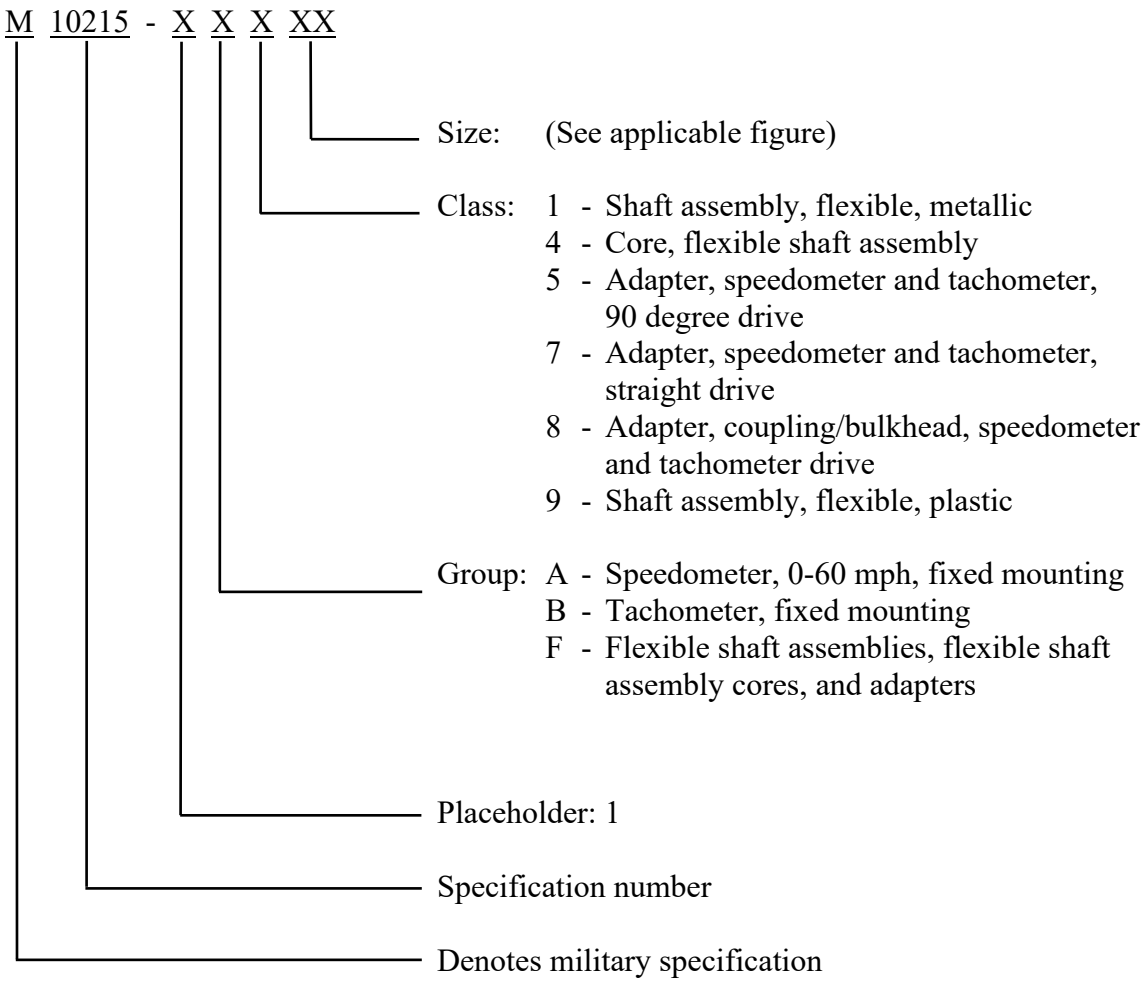
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1.2.2 Classes. The classes of shaft assemblies (flexible, metallic or plastic), cores and adapters are as follows:

Class 1 - Shaft assembly, flexible, metallic
Class 4 - Core, flexible shaft assembly
Class 5 - Adapter, speedometer and tachometer, 90 degree drive
Class 7 - Adapter, speedometer and tachometer, straight drive
Class 8 - Adapter, coupling/bulkhead, speedometer and tachometer drive
Class 9 - Shaft assembly, flexible, plastic

1.3 Part or identifying number (PIN). The PIN to be used for instruments acquired to this specification is created as follows:



2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

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2.2 Government documents.

2.2.1 Specifications, and standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

FEDERAL SPECIFICATIONS

L-P-380	-	Plastic Molding Material Methacrylate
TT-C-490	-	Chemical Conversion Coatings and Pretreatments for Metallic Substrates (Base for Organic Coatings)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-W-80	-	Window, Observation, Acrylic Base, Antielectrostatic, Transparent (for Indicating Instrument)
MIL-PRF-22750	-	Coating, Epoxy, High-Solids
MIL-PRF-23827	-	Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-DTL-53022	-	Primer, Epoxy Coating, Corrosion Inhibiting Lead and Chromate Free
MIL-DTL-53030	-	Primer Coating, Epoxy, Water Based, Lead and Chromate Free
MIL-DTL-53039	-	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant
MIL-DTL-53072	-	Chemical Agent Resistant Coating (CARC) System Application Procedures and Quality Control Inspection
MIL-DTL-64159	-	Camouflage Coating, Water Dispersible Aliphatic Polyurethane, Chemical Agent Resistant

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-202-201	-	Method 201, Vibration
MIL-STD-202-213	-	Method 213, Shock (Specified Pulse)
MIL-STD-810	-	Test Method Standard - Environmental Engineering Considerations and Laboratory Tests
MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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STANDARDIZATION DOCUMENT

SD-6 - Provisions Governing Qualification

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ H1607331 - Zero Acceptance Number Sampling Plans

(Copies of this document are available at https://www.asq.org.)

ASTM INTERNATIONAL

ASTM A380/A380M	-	Standard Practice for Cleaning, Descaling, and Passivation of Stainless Steel Parts, Equipment, and Systems
ASTM B117	-	Standard Practice for Operating Salt Spray (Fog) Apparatus.
ASTM D3935	-	Standard Specification for Polycarbonate (PC) Unfilled and Reinforced Material
ASTM G21	-	Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi

(Copies of these documents are available online at https://www.astm.org.)

SAE INTERNATIONAL

SAE AMS-STD-595 - Colors Used in Government Procurement

(Copies of these documents are available online at https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Items furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list (QPL) before contract award (see 4.3 and 6.3).

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3.2 Materials. Materials shall be in accordance with figure 1 through figure 9 where described. Dials shall be fabricated from metal or metal alloys, and case windows shall be fabricated from clear plastic conforming to MIL-W-80 or, if molded, conforming to type I, class 3 of L-P-380 or group 1, class 1 of ASTM D3935. For all other components, unless specified herein, materials shall be in accordance with the manufacturer's standard practice.

3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Design and construction. The design and construction of the instruments and components shall be in accordance with the interface and envelope dimensions in figure 1 through figure 9; however, the figures are not intended to limit the manufacturer's design.

3.3.1 Speedometer and tachometer. The instruments shall be provided with a bezel, a means for mounting and a connector for attachment of a flexible drive shaft assembly. A case window shall be provided to facilitate illumination of the dial from an outside source, and a means shall be provided to dehumidify trapped air after the speedometer or tachometer is sealed. The dehumidification component shall be secured so it will not become loose when the speedometer or tachometer is subjected to vibration.

3.3.1.1 Dial.

3.3.1.1.1 Marking. The face of the dial shall include the markings specified in figure 8, figure 9 and table I. The printing shall be commercial style lettering of the size specified in table I. The dial face shall have a black background with white numerals, letters, and graduations conforming to SAE AMS-STD-595.

TABLE I. Dial face marking.

Marking	Height (inches)	Height (mm)
Numerals for mph scale	0.219	5.6
Graduations, major, mph scale	0.219	5.6
Graduations, minor, mph scale	0.125	3.2
"Miles per hour" or "MPH"	0.125	3.2
"Revolutions per minute" or "RPM"	0.125	3.2
"Hours" and "10th" or "100th"	0.094	2.4
"Miles" and "10th"	0.094	2.4
MS or military part number	0.094	2.4
Manufacturer's identification	0.063 (max.)	1.6

3.3.1.1.2 Fastening. The dial shall be secured to the body of the speedometer or tachometer in such a position that the marking on the face of the dial and the numerals on the odometer or engine hour meter shall be legible when illuminated through the case window from an outside source.

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3.3.1.2 Pointer. The pointer shall be attached to the indicator movement and shall be damped to reduce oscillation from vibration. The pointer shall not touch the dial or window. The shape of the pointer shall facilitate reading of any point in the range of the dial. Not less than the top half of the visible portion of the pointer shall be white conforming to SAE AMS-STD-595. Damping of oscillation shall be in accordance with table II.

3.3.1.3 Recording devices.

3.3.1.3.1 Odometer. The instrument, when furnished for use with a speedometer, shall be of the non-reset-type, consisting of six numeral wheels located in the dial as shown in figure 8. The odometer shall totalize and record actual mileage up to 99,999.9 in multiples of 0.1 mile before automatically turning to zero. Accuracy shall be in accordance with table II.

3.3.1.3.2 Engine hour meter. The engine hour meter shall accurately record hours of engine operation. It shall totalize and record up to 9,999.9 hours in multiples of 0.1 hour or up to 9,999.99 hours in multiples of 0.01 hour before automatically turning to zero. Accuracy shall be in accordance with table II.

3.3.2 Drive mechanism.

3.3.2.1 Flexible shaft assembly. A flexible shaft assembly shall drive speedometers and tachometers with adapters as required. The assembly shall consist of a flexible outer casing and a flexible inner core. The outer casing shall be equipped with reusable means for attaching the assembly to both the instrument and the take-off point. The configuration shall fall within the envelope limits indicated in figure 1, figure 2 or figure 3, as applicable. End connections affecting interchangeability shall conform to the interface requirements in figure 1, figure 2 and figure 3. A means shall be provided in the assembly to prevent the core from falling out.

3.3.2.1.1 Core. The flexible shaft assembly core shall conform to figure 4. The outer layer of core wires shall be wound in the direction of right hand lay.

3.3.2.2 Adapters. Adapters shall conform to figure 5 (figure 5 cont.), figure 6 or figure 7, as applicable.

3.4 Performance. Table II applies. Verification of the parts under consideration will be accomplished under the laboratory conditions in 4.2.1 and tested as specified in 4.5.2. All of the components described by this specification shall meet the minimum requirements of table II after being tested under a variety of operational conditions to include: a) resistance to wear, b) verification of operating torque for shaft assemblies and adapters to include flexibility, torsional and tensile strength and c) resistance to potentially damaging environmentally induced conditions to include lubrication leakage, vibration resistance, shock resistance, water submersion, corrosive conditions, fungus contamination and ozone exposure (see table III).

3.4.1 Speedometer. When tested as specified in 4.5.2.1, the speedometer shall accurately indicate the vehicle speed when the drive shaft is rotated at the specified speeds.

3.4.2 Tachometer. When tested as specified in 4.5.2.2, the tachometer pointers shall accurately indicate a rotating speed on the dial twice that of the drive shaft speed.

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3.4.3 Odometer. When tested as specified in 4.5.2.3, the odometer-indicated mileage shall be compared with the actual mileage.

3.4.4 Engine hour meter. When tested as specified in 4.5.2.4, engine operation shall be recorded in hours by the engine hour meter on the tachometer.

3.4.5 Flexible shaft assembly, core assembly and adapter. When tested as specified in 4.5.2.5, the speedometer's pointer shall be observed.

3.5 Painting. Unless otherwise specified (see 6.2), cleaning, pretreatment, priming and painting shall be CARC in accordance with MIL-DTL-53072. When instrument bezels made of stainless steel are used, they shall be passivated in accordance with ASTM A380/A380M, pretreated in accordance with TT-C-490, primed in accordance with MIL-DTL-53022 Type IV or MIL-DTL-53030 Type II (dry film thickness of 1.5-2.5 mils), and top coated in accordance with MIL-DTL-64159 or MIL-DTL-53039, color Green 383 (Chip 34094) in accordance with SAE AMS-STD-595, to a dry film thickness of 2.0 to 3.0 mils. When speedometers and tachometers (figure 8 and figure 9) are used in the interior of a vehicle or system, they shall be cleaned and pretreated in accordance with TT-C-490, primed in accordance with MIL-DTL-53022 Type IV or MIL-DTL-53030 Type II (dry film thickness of 1.5-2.5 mils), and top coated with an epoxy topcoat in accordance with MIL-PRF-22750, colors Green 383 (Chip 34094) or Field Green (Chip 34095) in accordance with SAE AMS-STD-595, to a dry film thickness of 1.7 to 2.3 mils.

3.6 Identification and marking. Unless otherwise specified (see 6.2), identification and marking of the items shall be in accordance with MIL-STD-130. As a minimum, it shall consist of the manufacturer's CAGE code and the applicable military part number. All markings shall be permanently and legibly printed, labeled, or stamped on the dial of the speedometers and tachometers, on the casings of the flexible shaft assemblies and on the housing of the adapters. The drive ratio and date of manufacture shall also be marked on the housing of all adapters, and the flexible shaft assembly cores shall be tagged.

3.7 Workmanship. Components shall be free of defects such as rust, burrs and sharp edges that may result in injury of personnel (see 4.5.1). All solder cleaning agents and flux shall be non-corrosive. All traces of flux shall be removed from soldering.

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TABLE II. Performance requirements.

Part	Requirements				Verification
Speedometer 3.4.1	Drive shaft speed (rpm)	Observed readings at =>	77 ºF	160 ºF	4.5.2.1
		(mph)			
	167	10	+1 to -1	+4 to -4	
	500	30	+2 to -1	+5 to -4	
	1,000	60	+3 to -0	+6 to -3	
	Pointers shall leave the stop at or before the drive shaft reaches a speed that equates to 3 mph				
	Pointers shall be damped to prevent an oscillation of more than ±0.5 mph above 10 mph when operated directly off a drive shaft with no fluctuation in speed				
Tachometer 3.4.2	Drive shaft speed (rpm)	Observed readings at =>	77 ºF	160 ºF	4.5.2.2
		(rpm)			
	250	500	±60	±110	
	1000	2000	±60	±110	
	1500	3000	±60	±110	
	Pointers shall leave the stop at or before the drive shaft reaches a speed that equates to 100 rpm				
	Pointers shall be damped to prevent an oscillation of more than ±30 rpm above 500 rpm when operated directly off a drive shaft with no fluctuation in speed				
Odometer 3.4.3	Readings shall be accurate within ± 1% of actual mileage				4.5.2.3
Engine Hour Meter 3.4.4	Readings shall be accurate within ± 2% of actual elapsed time				4.5.2.4
Flexible shaft assembly, core assembly and adapter 3.4.5	10 minutes of operation at 35 mph during which time the speedometer's pointer shall not exceed ±2 mph in oscillation				4.5.2.5

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspections (see 4.3).
b. Conformance inspections (also known as Acceptance inspections) (see 4.4).

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions.

4.2.1 Conditions. Tests shall be conducted under general laboratory conditions acceptable to the Federal Government.

4.2.1.1 Temperatures. Unless otherwise specified herein, tests and temperature tolerances shall be conducted at an ambient temperature of 77 °F ± 18 °F (25 °C ± 10 °C), hereafter referred to as simply 77 °F. If specified, some high temperature tests will be conducted at 160 °F ± 5 °F (71 °C ± 3 °C), hereafter referred to as simply 160 °F while low temperature tests, if specified, will be conducted at -65 °F ± 5 °F (-54 °C ± 3 °C), hereafter referred to as simply -65 °F. Unless otherwise specified herein, the test specimen shall be thermally stabilized for 1 hour prior to being tested. Note: The specified temperature tolerances apply throughout this document.

4.2.1.2 Pressure. Unless otherwise specified herein, all qualification tests shall be conducted at a pressure of 101.3 kPa ± 5 percent.

4.2.1.3 Humidity. Unless otherwise specified herein, all qualification tests shall be conducted at a humidity level of 50 percent ± 5 percent.

4.2.2 Standard specimens. Unless otherwise specified (see 6.2), and excepting acceptance tests, the shaft assembly shall conform to size no. 05 of figure 1 or size no. 05 of figure 2.

4.2.3 Apparatus. In addition to the standard measuring equipment, the test apparatus shall include a temperature, pressure and humidity controlled air chamber (if required); a submersion tank; and a test stand with suitable drives for rotating the drive shaft at controlled speeds. Examinations requiring magnification shall be conducted using a lens with a minimum of 5X magnification (typical of a hand-held lens) hereafter simply referred to as a 5X lens.

4.2.3.1 Saline solution. A saline solution shall be used for the vacuum and pressure tests in 4.5.5.4.1. The salt used shall be sodium chloride containing on a dry basis not more than 0.1 percent of sodium iodide and not more than 0.2 percent of total impurities. The solution shall be prepared by dissolving 5 parts by weight of salt in 95 parts by weight of distilled water and shall be kept free of sediment by filtration or decantation.

4.3 Qualification inspections. Qualification inspections (see 3.1) shall include examination of the product (see 4.5.1) and additional tests from table III as applicable, followed by verification of performance (see 3.4 and table II).

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TABLE III. Classification of inspections.

Title	Requirement	Inspection	
		Qualification[1]	Conformance[1]
Examination of the product	3.2, 3.3, 3.5, 3.6, 3.7	4.5.1	4.5.1
Performance	3.4	4.5.2	4.5.2
Operational Testing	3.4	4.5.2	4.4
Resistance to wear		4.5.3	
Operating torque		4.5.4	4.5.4.4 [1]
Flexible shaft assembly		4.5.4.1	
Flexibility		4.5.4.2[2]	4.5.4.2
Adapter		4.5.4.3	
Torsional and tensile strengths		4.5.4.4[3]	4.5.4.4[3]
Environmental		4.5.5	
Lubrication leakage		4.5.5.1	
Vibration resistance		4.5.5.2	
Shock resistance		4.5.5.3	
Waterproofness		4.5.5.4	4.5.5.4.2[4]
Corrosion resistance		4.5.5.5	
Fungus resistance		4.5.5.6	
Ozone resistance		4.5.5.7	

[1] Only the testing in 4.5.4.4 (for flexible shaft assemblies) is required.
[12] Tests may not be applicable to all groups or classes. See tables IV and V for a more detailed breakdown of which tests are required. Applies to non-metallic shaft assemblies only.
[3] Applies to flexible shaft assemblies only.
[4] Applies to flexible shaft assemblies and adapters only.

4.3.1 Qualification inspections. Qualification inspection sample item and testing sequence shall be in accordance with table IV. Two sample items will be tested through each sequence (i.e., sequence A has sample items A-1 and A-2, each of which will be subjected to the required Examination of Product inspection, followed by two additional tests, the Resistance to Wear and the Waterproofness test. The required Performance inspections for each sample item complete sequence A).

4.3.2 Maintenance of qualification. At specified intervals determined by the qualifying activity, the manufacturer must be able to demonstrate the company still has the capabilities and facilities necessary to produce the QPL items in accordance with this specification and in accordance with the provisions governing qualification in SD-6.

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Table IV. Sequence of qualification inspections and testing.

Sample Item	Initial Test	Additional Testing - Test Sequence			Final Test
		Test #1	Test #2	Test #3	
Speedometers & Tachometers					
A-1 A-2	Examination of Product 4.5.1	Resistance to wear 4.5.3.1, table VII	Waterproofness 4.5.5.4.1		Performance 4.5.2
B-1 B-2	Examination of Product 4.5.1	Vibration 4.5.5.2	Corrosion resistance 4.5.5.5		Performance 4.5.2
C-1 C-2	Examination of Product 4.5.1	Shock resistance 4.5.5.3	Fungus resistance 4.5.5.6		Performance 4.5.2
D-1 D-2	Examination of Product 4.5.1	Waterproofness 4.5.5.4.1, 4.5.5.4.3	No further testing Required		
Flexible shaft assemblies					
E-1 E-2	Examination of Product 4.5.1	Resistance to wear 4.5.3.2, table VIII	Waterproofness 4.5.5.4.2	Corrosion resistance 4.5.5.5	Performance 4.5.2
F-1 F-2	Examination of Product 4.5.1	Lubrication leakage 4.5.5.1	Operating torque 4.5.4.1	Waterproofness 4.5.5.4.2	Performance 4.5.2
G-1 G-2	Examination of Product 4.5.1	Flexibility (non-metallic only) 4.5.4.2	Oscillation 4.5.3.2.1	Fungus resistance 4.5.5.6	Performance 4.5.2
H-1 H-2	Examination of Product 4.5.1	Torsional and tensile strengths 4.5.4.4	Ozone resistance 4.5.5.7		Performance 4.5.2
I-1 I-2	Examination of Product 4.5.1	Waterproofness 4.5.5.4.2, 4.5.5.4.3	No further testing Required		
Adapters					
J-1 J-2	Examination of Product 4.5.1	Resistance to wear 4.5.3.3, table_VIII	Waterproofness 4.5.5.4.2		Performance 4.5.2
K-1 K-2	Examination of Product 4.5.1	Operating torque 4.5.4.3			Performance 4.5.2
L-1 L-2	Examination of Product 4.5.1	Waterproofness 4.5.5.4.2, 4.5.5.4.3	No further testing Required		
M-1 M-2	Examination of Product 4.5.1	Corrosion resistance 4.5.5.5			Performance 4.5.2
N-1 N-2	Examination of Product 4.5.1	Fungus resistance 4.5.5.6			Performance 4.5.2
Flexible shaft assembly cores					
O-1 O-2	Examination of Product 4.5.1	Torsional and tensile strengths 4.5.4.4	No further testing Required		

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4.4 Conformance inspection. Conformance inspection testing (also known as Acceptance inspection testing) and test sequence is described in table V and shall include the examination of the product (see 4.5.1) and the applicable requirements and tests of table III. Each production lot (see 4.4.1.1) shall include conformance testing as defined herein.

4.4.1 Sampling plan. Unless otherwise specified in 6.2, sampling procedures for conformance inspection examinations and tests shall be from the inspection lot. Sampling procedures for performance testing shall be in accordance with MIL-STD-1916, verification level I or ASQ H1607, AQL 10.0 and sampling for all other examinations and tests shall be in accordance with MIL-STD-1916, verification level II or ASQ H1607331, AQL 4.0.

4.4.1.1 Lot formation. An inspection lot shall consist of all speedometers, tachometers, flexible shaft assemblies or adapters of a single group, class and PIN, from an identifiable production period, from one manufacturing location submitted at the same time for acceptance.

TABLE V. Sequence of conformance inspections and testing.

Item type	Initial Test	Additional Testing - Test Sequence		Final Test
		Test #1	Test #2	
Speedometers (Group A) & Tachometers (Group B)	Examination of Product 4.5.1			Performance 4.5.2
Flexible Shaft Assemblies (Classes 1 and 9)	Examination of Product 4.5.1	Operating Torque 4.5.4.2[1], 4.5.4.4	Waterproofness 4.5.5.4.2	Performance 4.5.2
Flexible Shaft Assembly Cores (Class 4)	Examination of Product 4.5.1			Operating Torque 4.5.4.4
Adapters (Classes 5, 7, and 8)	Examination of Product 4.5.1		Waterproofness 4.5.5.4.2	Performance 4.5.2

[1] Applies to non-metallic shaft assemblies only.

4.5 Methods of inspection.

4.5.1 Examination of product. Instruments and components shall be examined for the defects listed in table VI. Examination shall be visual, tactile or by measurement with standard inspection equipment.

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TABLE VI. Examination of defects.

Defect	Method of examination
Design, materials and construction affecting interchangeability (see 3.2, 3.3)	Standard inspection equipment and functional
Design, materials and construction not affecting interchangeability (see 3.2, 3.3)	Standard inspection equipment and functional
Painting not as specified (see 3.5)	Visual
Identification and marking not as specified (see 3.6)	Visual
Workmanship (see 3.7)	Visual and tactile

4.5.2 Performance testing. All of the components described by this specification shall meet the minimum performance requirements even after being subjected to a variety of operational conditions as demonstrated through a series of tests (see table III): a) resistance to wear testing, b) testing of operating torque for shaft assemblies and adapters to include flexibility, torsional and tensile strength and c) environmental testing to include lubrication leakage, vibration resistance, shock resistance, waterproofness, corrosion, fungus and ozone testing. At the conclusion of each required testing sequence, all samples (except those disassembled for inspection) shall meet the performance requirements of 3.4 and table II.

4.5.2.1 Speedometer. Speedometers shall be mounted in an air chamber with the dial face tilted 10 degrees backward from the vertical. while connected to a flexible shaft assembly, The air chamber shall be maintained at each of the following temperatures for 4 hours: 77 °F and 160 °F (conformance testing will be conducted at an ambient air temperature of 77 °F only). After each 4-hour period and while still maintaining the temperature, the drive shaft shall be rotated at 167 rpm, 500 rpm, and 1,000 rpm. , and tThe speedometer shall meet the performance requirements of 3.4.1 whenand the pointer indications areshall be compared to the requirements in table II.

4.5.2.1.1 Conformance testing of speedometers. Conformance testing of speedometers shall be conducted using the procedure described in 4.5.2.1 however testing shall only be conducted at an ambient air temperature of 77 °F. Stabilizing for 4 hours is not required provided the speedometers have been in ambient conditions for at least that long.

4.5.2.2 Tachometer. Tachometers shall be mounted in an air chamber with the dial face tilted 10 degrees backward from the vertical. The air chamber shall be maintained at each of the following temperatures for 4 hours: 77 °F and 160 °F. After each 4-hour period and while still maintaining the temperature, the drive shaft shall be rotated at 250 rpm, 1,000 rpm, and 1,500 rpm. The tachometer shall meet the performance requirements of 3.4.2 when the pointer indications are compared to the requirements in table II.Tachometers shall be tested in a manner similar to speedometers as specified in 4.5.2.1., except that the drive shaft shall be rotated at 250 rpm, 1,000 rpm, and 1,500 rpm and shall meet the performance requirements set forth in 3.4.2 The tests shall be conducted with the air chamber temperature stabilized at 77 ºF and 160 ºF (conformance testing will be conducted at an ambient air temperature of 77 °F only) and the results compared to the requirements in table II.

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4.5.2.1.1 Conformance testing of tachometers. Conformance testing of tachometers shall be conducted using the procedure described in 4.5.2.2 however testing shall only be conducted at an ambient air temperature of 77 °F. Stabilizing for 4 hours is not required provided the tachometers have been in ambient conditions for at least that long.

4.5.2.3 Odometer (speedometers only). The drive shaft shall be rotated for a continuous period of 6 hours at a speed of 1,000 rpm at 77 °F at the conclusion of the speedometer test specified in 4.5.2.1. The odometer-indicated mileage for the 6-hour period shall be compared with the actual calibrated mileage and with the requirements in table II.

4.5.2.4 Engine hour meter (tachometers only). The drive shaft shall be rotated at 102600 rpm for 6 hours at the conclusion of the tachometer test specified in 4.5.2.2. During and after the test, the engine hour meter reading shall be observed. The meter readings shall be compared with the actual run times, and the results shall comply with the requirements in table II.

4.5.2.5 Flexible shaft assembly, core assembly, and adapter. The flexible shaft assemblies, core assemblies, and adapters shall be connected to a previously calibrated speedometer and shall be rotated to indicate 35 mph for a period of 10 minutes. The speedometer shall be observed for any deviations from the requirements of table II.

4.5.3 Resistance to wear testing.

4.5.3.1 Speedometer and tachometer. Speedometers and tachometers shall be operated for 1,000 hours at the speeds and temperatures specified in table VII without failure.

TABLE VII. Speedometer and tachometer wear test.

Hours	Temperature	Speed
750	77 °F	Equal indicating periods for 1/3, 2/3 and full scale readings
250	160 °F	(Same as above)

4.5.3.2 Flexible shaft assembly. Flexible shaft assemblies 10 feet (ft) (3.05 meters (m)) long shall be mounted in a test fixture in such a manner that the shaft is confined in a S-shape having two bends of 180 degrees in the same plane, with each bend having a 5-inch (12.7-centimeter (cm)) radius. With a constant load of 22 ± 2 ounce-force inches (ozf inches), (0.16 ± 0.01 Newton meters (Nm)) at the speedometer and tachometer end of the shaft core, the core shall be operated at the speeds, temperatures, and times specified in table VIII without failure.

TABLE VIII. Flexible shaft assembly and adapter wear test.

Hours	Temperature	Speed (rpm)
100	77 °F	750
100	77 °F	1,250
200	77 °F	2,000
25	160 °F	750

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25	160 °F	1,250
50	160 °F	2,000

4.5.3.2.1 Flexible shaft assembly, oscillation. Flexible shaft assemblies 10 feet (3.05 m) long shall be connected to a previously calibrated speedometer and shall be rotated at indicated speedometer speeds from 10 to 60 mph at 10 mph intervals. Running time at each speed shall then be repeated with a previously calibrated tachometer. The indicated tachometer rpm shall be between 500 and 4,000 at 500 rpm intervals with a running time of 10 minutes at each rotating speed. Pointer oscillation shall not exceed 1 mph at speeds above 10 mph on the speedometer dial and shall not exceed 50 rpm on the tachometer dial at rotating speeds above 500 rpm.

4.5.3.3 Adapter. Adapters shall be mounted in a fixture with a constant load of 22 ± 2 ozf inches (0.16 ± 0.01 Nm) and operated at the speeds, temperatures and times specified in table VIII without failure.

4.5.4 Operating torque testing.

4.5.4.1 Flexible shaft assembly. Flexible shaft assemblies (see 4.2.2) shall be placed in a test fixture with the test sample temperature and air temperature maintained constant at -65 °F for 7 hours. At the conclusion of the 7-hour period and while still at the above specified temperature, the core shall be rotated without load to produce a starting and uniformly acceleratingon from 0 to 1,000 rpm in 15 seconds. The torque required to rotate the core shall be measured and shall at no time exceed 80 ozf inches (0.56 Nm).

4.5.4.2 Flexibility (non-metallic shaft assemblies only). A section of flexible shaft assembly approximately 17 inches (43 cm) from one end shall be bent 180 degrees around a 10-inch (25.4-cm) diameter mandrel in an ambient air temperature of 77 °F. A spring scale or other suitable measuring device shall be used at the fitting end to record the maximum force required. The applied force required shall be no greater than 8 pounds-force (lbf) (35.6 Newtons (N)).

4.5.4.3 Adapter. Adapters shall be placed in a test fixture with the test sample temperature and air temperature maintained constant at -65 °F for 7 hours. At the conclusion of the 7-hour period and while still held at the above specified temperature, the core shall be rotated without load to produce a uniform acceleration from 0 to 1,000 rpm in 15 seconds. The torque required to rotate the core shall be measured and shall at no time exceed 80 ozf inches (0.56 Nm).. The torque required to rotate the core in starting and accelerating shall be measured and while uniformly accelerating from 0 to 1,000 rpm in 15 seconds shall at no time exceed 80 ozf inches (0.56 Nm).

4.5.4.4 Torsional and tensile strengths (flexible shaft assemblies and cores only). Flexible shaft assembly cores shall be placed in a torsion-testing machine in a straight position without slack so that 10 inches (25.4 cm) of the speedometer and tachometer end of the core will be subjected to torsion. The cores shall be twisted in a clockwise direction. The test shall be repeated with the drive end of the core in the machine. The core end fittings shall then be subjected to a load applied in tension to the core with the fittings anchored. In each instance, the end fittings of the core assembly shall withstand torsional breaking force of the cores shall be not less than 15 lbf inches (1.7 Nm) in torsion, and the end fittings attached to the core shall withstand a load of not less than 50 lbs (22.7 kg) when applied in tension, without exhibiting any plastic deformation.

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4.5.5 Environmental testing.

4.5.5.1 Lubrication leakage (lubricated flexible shaft assembliesy only). Flexible shaft assemblies shall be carefully sealed at both ends. Oil, grease, and dirt should be removed by degreasing the assembly and sealed ends with a solvent cleaner and scrubbing and then thoroughly drying. Flexible shafts shall then be placed in a temperature controlled air chamber at a constantly maintained temperature of 250 °F ± 5 °F (121 °C ± 3 °C) for 2 hours. At the conclusion of the 2-hour period, each assembly shall be removed from the air chamber and suspended from the transmission end in a vertical position. The lower end of each flexible shaft assembly shall be wiped over the entire area with a good quality, absorbent tissue. Then the tissue shall be visually inspected; no evidence of absorbed lubricant shall be allowed. Subsequently, the flexible shaft assembly shall be subjected to the test specified in 4.5.4.1

4.5.5.2 Vibration resistance (speedometers and tachometers only). Speedometers and tachometers shall be mounted as if in operation and subjected to the vibration test as specified in MIL-STD-202-201, except that motion shall be applied for 1 hour in each direction of the three major axes. During vibration, each speedometer and tachometer shall be operated in such a manner that the speedometer and tachometer pointer shall be at midscale, and observation shall be made for oscillation. At the conclusion of the test, neither the components nor the assemblies shall show any evidence of cracking or disintegration.

4.5.5.3 Shock resistance (speedometers and tachometers only). Speedometers and tachometers shall be mounted as if in operation and subjected to the shock test as specified in MIL-STD-202- 213, using the saw-tooth shock pulse procedure. The test shall be repeated 10 times and applied in the vertical axis. During shock, the speedometer and tachometer shall be operated in such a manner that the speedometer and tachometer pointer shall be at midscale, and observation shall be made for oscillation. At the conclusion of the test, they shall show no evidence of pointer sticking, no delay of functioning and no disintegration of component materials or assemblies.

4.5.5.4 Waterproofness testing (- Sspeedometers, tachometers, flexible shaft assemblies and adapters only). When the speedometers, tachometers, flexible shaft assemblies, and adapters are tested in accordance with the requirements in 4.5.5.4.1 through 4.5.5.4.3, they shall show no evidence of leakage. When called out, the saline solution shall be in accordance with 4.2.3.1.

4.5.5.4.1 Vacuum and pressure tests (for speedometers and tachometers only).

4.5.5.4.1.1 External vacuum test. Speedometers and tachometers shall be subjected to an external vacuum test to verify waterproofness. The speedometer and tachometer connectors shall be sealed with caps simulating the mating flexible cable connectors or as otherwise specified. The flexible shaft assemblies and adapters shall be sealed against leakage. The test chamber shall be evacuated to a pressure 6 pounds below atmospheric pressure to create a minimum of 6 pounds per square inch (psi) (0.42 kilogram-force per centimeter square ($kg(f)/cm^2$)) of internal pressure to all voids within the component while totally immersed in the saline solution (see 4.2.3.1) for 30 minutes during qualification testing. The component shall be carefully observed for air bubbles from the interior of the device during the specified time period. Bubbles that are the result of entrapped air on the various exterior surfaces of the device shall not be considered a leak.

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4.5.5.4.1.2 External pressure test. Following the external vacuum test in 4.5.5.4.1.1, the speedometers and tachometers shall be subjected to an external pressure test to verify waterproofness. The speedometer and tachometer connectors shall be sealed with caps simulating the mating flexible cable connectors. The test chamber shall then be pressurized to 6 pounds above atmospheric to create a minimum of 6 psi (0.42 kg (f)/cm^2) of internal pressure to all voids within the component while totally immersed in the saline solution (see 4.2.3.1) for 30 minutes during qualification testing. The component shall be carefully observed for air bubbles from the interior of the device during the specified time period. Bubbles that are the result of entrapped air on the various exterior surfaces of the device shall not be considered a leak.

4.5.5.4.2 Internal pressure test for (flexible shaft assemblies and adapters only). Finally, tThe flexible shaft assemblies and adapters shall be subjected to an internal pressure test to verify waterproofness. After sealing the flexible shaft assemblies and adapters against leakage, the test chamber shall then be pressurized to 6 pounds above atmospheric to create 6 psi (0.42 kg(f)/cm^2) of internal pressure while totally immersed in tap water for 30 minutes during qualification testing and 5 minutes during conformance testing. The component shall be carefully observed for air bubbles from the interior of the device during the specified time period. Bubbles that are the result of entrapped air on the various exterior surfaces of the device shall not be considered a leak.

4.5.5.4.3 Post inspection (for speedometers, tachometers, and flexible shaft assemblies, and adapters only). Following the application of the external and internal pressure tests, a selected portion of the test components shall be disassembled and inspected for the presence of water (see table IV). If water is present, the component shall be rejected.

4.5.5.5 Corrosion resistance testing - S(speedometers, tachometers, flexible shaft and core assembliesy, and adapters only). Speedometers, tachometers, flexible shafts and core assemblies, and adapters shall be tested for corrosion resistance test by means of the salt spray (fog) test as specified in ASTM B117. The speedometers, tachometers and both ends of the flexible shaft assemblies and adapters shall be sealed against leakage with connectors simulating the applicable mating parts or as otherwise specified, and the test shall be run in four alternating 24-hour periods of salt fog exposure and drying conditions as specified in Method 509 of MIL-STD-810. After testing, each sample part shall be examined using a 5X lens. There shall be no evidence of corrosion such as pitting or cracking causing impairment of operation, and each sample part shall operate satisfactorily as required by 3.4. Material discoloration which does not affect performance is not necessarily disqualifying; however, any and all evidence of corrosion shall be reported in the test report.

4.5.5.6 Fungus resistance testing -– (sSpeedometers, tachometers, flexible shaft and core assemblyies, and adapters only). Speedometers, tachometers, flexible shaft and core assemblies, and adapters shall be tested as specified in ASTM G21. The speedometers and tachometers and both ends of the flexible shaft assemblies and adapters shall be sealed against leakage with connectors simulating the applicable mating parts or as otherwise specified, and the test shall be run for 28 days as specified in ASTM G21 and Method 508 of MIL-STD-810. After testing, each sample shall be examined using 5X lens. There shall be no evidence of fungus or microbial growth such as discoloration; noticeable, abnormal growth or other similar evidence, and each sample part shall operate satisfactorily as required by 3.4. Any and all evidence of fungus or microbial growth shall be reported in the test report.

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4.5.5.7 Ozone resistance testing (- Fflexible shaft assembliesy only). Flexible shaft assemblies shall be conditioned at ambient room temperature for 45 minutes. The assemblies shall then be placed in a test exposure chamber containing air mixed with ozone in the proportion of 50 ± 5 parts per 100 million parts of atmospheric air by volume for 168 hours at a temperature of 100 °F ± 5 °F (38 °C ± 3 °C). Immediately after the completion of the test period, each sample shall be examined using a 5X lens with the specimen coiled in a 12-inch (30-cm) inside diameter loop. The components shall show no evidence of cracking, rupture or other deterioration, and each sample part shall operate satisfactorily as required by 3.4.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The speedometers, tachometers, flexible shaft assemblies, flexible shaft assembly cores and adapters covered by this specification are intended for use in indicating speed in miles per hour or engine speed in revolutions per minute, and to record total distance in miles traveled by the vehicle or total hours of operation of the engine.

6.1.1 Military unique. The speedometers, tachometers, and related parts covered by this specification are intended for use in military motor vehicles. These devices may be subjected to a variety of environments and severe conditions including vibration, shock, saltwater submersion, and temperature extremes.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Group (see 1.2.1), class (see 1.2.2) and quantity required.
c. Painting, if other than as specified (see 3.5).
d. Identification and marking, if other than as specified (see 3.6).
e. Inspection conditions, if other than specified (see 4.2).
f. Standard specimens, if other than specified (see 4.2.2).
g. Sampling plan, if other than specified (see 4.4.1).
h. Packaging requirements (see 5.1).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in QPL-10215 whether or not such products have actually been listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Aviation VEB, 6090 Strathmore Rd8000 Jefferson Davis Highway, Richmond, VA 23237297-5616 or via email at STDZNMGT@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Subject term (key word) listing.

Engine
Mechanical
MPH
RPM
Vehicle

6.5 Changes from the previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.6 Supersession. This specification supersedes the documents listed in table IX.

Table IX. Supersession data.

Document	Date
MIL-PRF-10215J	16 January 2001
MIL-S-10215H	4 November 1992
MS35916D	11 July 1991
MS39021F	11 July 1991
MS39130D	14 November 1985
MS39132G	27 January 1988
MS51071H	12 August 1987
MS51072F	4 August 1987
MS52116A	19 January 1988
DS52140	28 September 1977
MS53099A	24 June 1970

6.7 Cross-reference data. Instruments and components conforming to this specification are interchangeable/substitutable with instruments and components conforming to the documents listed in table X.

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Table X. Cross-reference data.

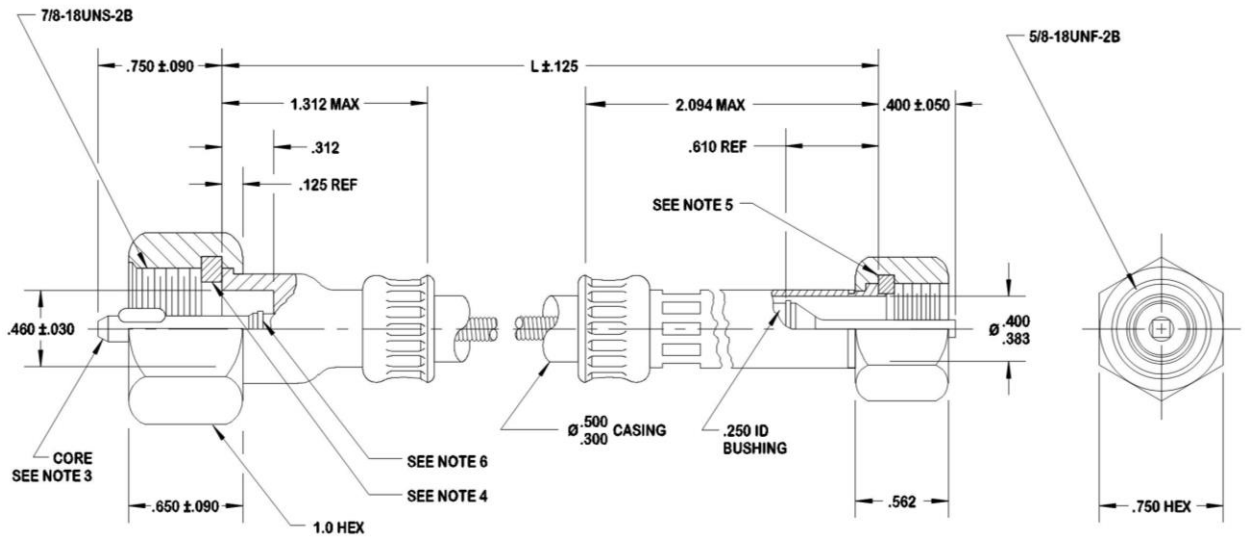
Document	Date	MIL-DTL-10215K	MIL-DTL-10215MP
MIL-PRF-10215J	16 January 2001	N/A	N/A
MIL-S-10215H	4 November 1992	N/A	N/A
MS35916D	11 July 1991	Figure 10	Figure 9
MS39021F	11 July 1991	Figure 8	Figure 8
MS39130D	14 November 1985	Figure 5	Figure 5
MS39132G	27 January 1988	Figure 6	Figure 6
MS51071H	12 August 1987	Figure 1	Figure 1
MS51072F	4 August 1987	Figure 4	Figure 4
MS52116A	19 January 1988	Figures 2 and 3	Figures 2 and 3
DS52140	28 September 1977	Figure 9	Deleted
MS53099A	24 June 1970	Figure 7	Figure 7

6.8 Cross-reference classification data. Table XI displays the revised classification data between MIL-PRF-10215J and MIL-DTL-10215M.

Table XI. Cross-reference classification data.

MIL-PRF-10215J	MIL-DTL-10215K	MIL-DTL-10215MP
Type I	Type I	Type I
Type II	Deleted	Deleted
Group A	Group A - Speedometer (0-60 miles per hour), fixed mounting	Group A - Speedometer (0-60 miles per hour), fixed mounting
Group B	Group B - Tachometer, fixed mounting	Group B - Tachometer, fixed mounting
Group C	Deleted	Deleted
Group D	Deleted	Deleted
	Group E - Speedometer (0-100 k/hr), fixed mounting	Deleted
	Group F - Flexible shaft assemblies, flexible shaft core assemblies and adapters	Group F - Flexible shaft assemblies, flexible shaft core assemblies and adapters
Class 1	Class 1	Class 1
Class 2	Deleted	Deleted
Class 3	Deleted	Deleted
Class 4	Class 4	Class 4
Class 5	Class 5	Class 5
Class 7	Class 7	Class 7
Class 8	Class 8	Class 8
Class 9	Class 9	Class 9

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NOTES:
1. All dimensions are shown in inches.
2. Unless otherwise specified, all tolerances shall be ±.010.
3. For core assembly details and associated core assemblies, see figure 4.
4. For gasket material and dimensions, see Army part number 7539688.
5. For gasket material and dimensions, see Army part number 7539689.
6. For washer material and dimensions, see Army part number 5329467.
7. Casing shall be metal covered with rubber or plastic to meet performance requirements of procurement specification.

FIGURE 1. Shaft assembly, flexible metallic, tachometer and speedometer.

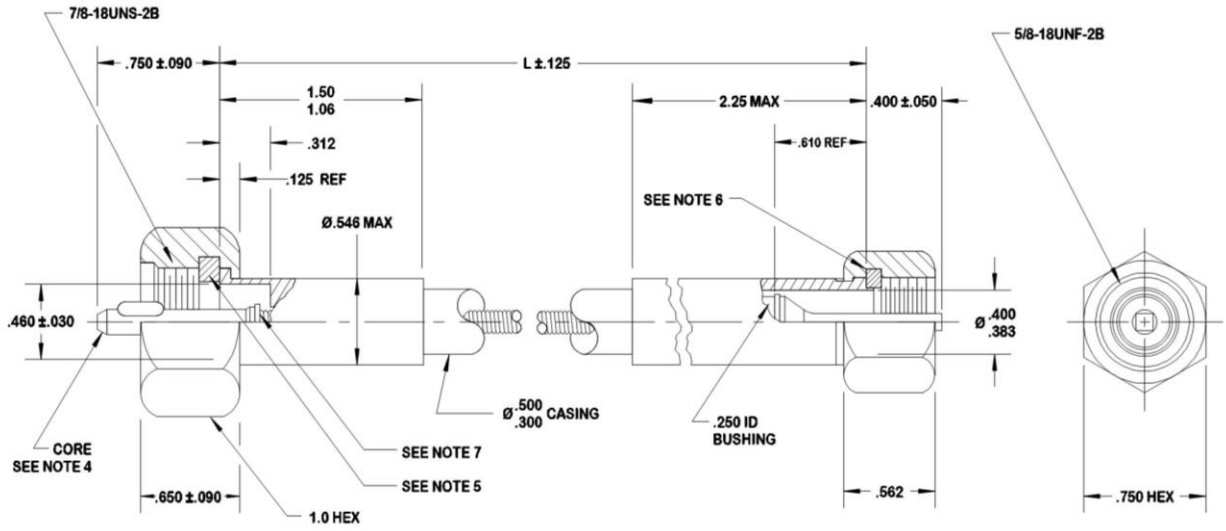
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Size no.	Former MS part no.	Former Army part no.	Shaft length
01	MS51071-1	8735736	17.5
02	MS51071-2	7976738	22.5
03	MS51071-3	7983060	40
04	MS51071-4	7527480	55
05	MS51071-5	7320561	62.5
07	MS51071-7	7064526	67.5
08	MS51071-8	7415912	72.5
09	MS51071-9	7096559	77.5
10	MS51071-10	7526039	85
11	MS51071-11	7412913	90
12	MS51071-12	7983059	95
13	MS51071-13	7527479	100
14	MS51071-14	7983213	107.5
15	MS51071-15	7983057	144
16	MS51071-16	7954735	156
17	MS51071-17	7987478	168
18	MS51071-18	8340482	216
19	MS51071-19	7344191	120
20	MS51071-20	8345795	138
21	MS51071-21	-	13
22	-	7952641	50.00
23	-	7389881	57.50
24	-	7356889	180.00

FIGURE 1. Shaft assembly, flexible metallic, tachometer and speedometer - Continued.

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PREFERRED CONSTRUCTION
(See figure 3 for optional construction)



NOTES:
1. All dimensions are shown in inches.
2. Unless otherwise specified, all tolerances shall be ±.010.
3. Preferred construction shown. For optional construction, see figure 3.
4. For core assembly details and associated core assemblies, see figure 4.
5. For gasket material and dimensions, see Army part number 7539688.
6. For gasket material and dimensions, see Army part number 7539689.
7. For washer material and dimensions, see Army part number 5329467.
8. Casing shall be metal covered with rubber or plastic to meet performance requirements of procurement specification.

FIGURE 2. Shaft assembly, flexible, plastic, tachometer and speedometer.

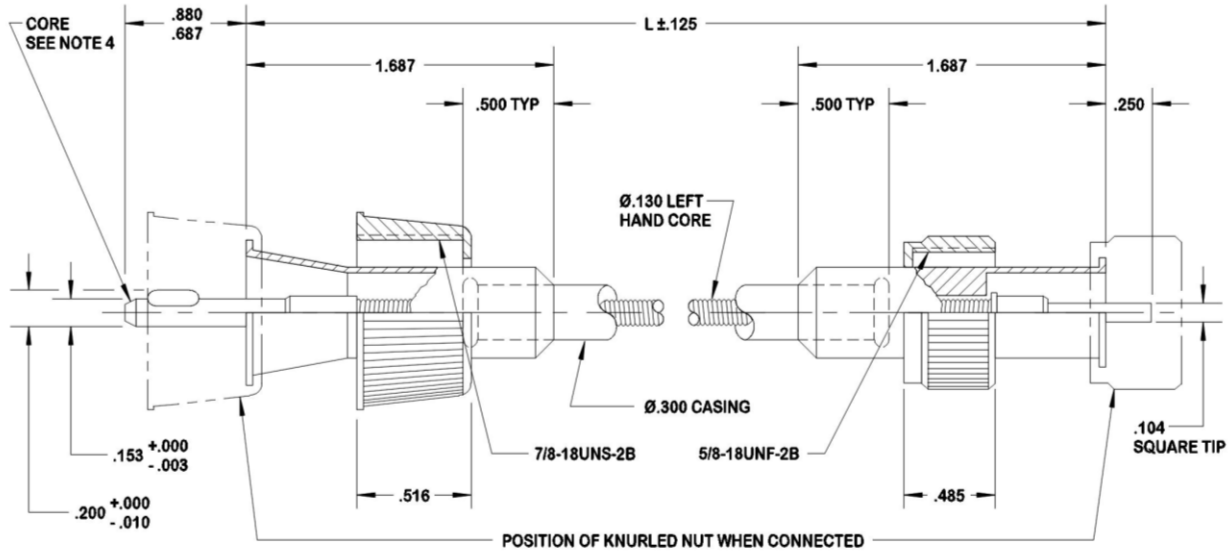
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Size no.	Former MS part no.	Shaft length
01	MS52116-1	17.5
02	MS52116-2	22.5
03	MS52116-3	40
04	MS52116-4	55
05	MS52116-5	62.5
07	MS52116-7	67.5
08	MS52116-8	72.5
09	MS52116-9	77.5
10	MS52116-10	85
11	MS52116-11	90
12	MS52116-12	95
13	MS52116-13	100
14	MS52116-14	107.5
15	MS52116-15	144
16	MS52116-16	156
17	MS52116-17	168
18	MS52116-18	216
19	MS52116-19	120
20	MS52116-20	138

FIGURE 2. Shaft assembly, flexible, plastic, tachometer and speedometer - Continued.

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OPTIONAL CONSTRUCTION
(See figure 2 for preferred construction)



NOTES:
1. All dimensions are shown in inches.
2. Unless otherwise specified, all tolerances shall be ±.010.
3. Optional construction shown. For preferred construction, see figure 2.
4. For core assembly details and associated core assemblies, see figure 4.
5. Casing shall be metal covered with rubber or plastic to meet performance requirements of procurement specification.

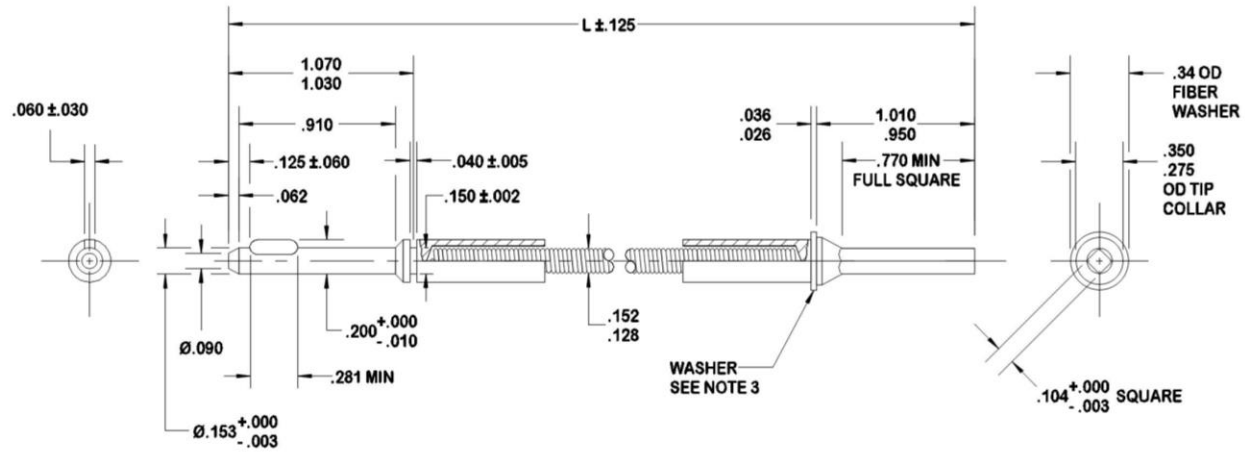
FIGURE 3. Shaft assembly, flexible, plastic, tachometer and speedometer - optional construction.

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Size no.	Former MS part no.	Shaft length
21	MS52116-21	72
22	MS52116-22	84
23	MS52116-23	96
24	MS52116-24	108
25	MS52116-25	120
26	MS52116-26	132
27	MS52116-27	144
28	MS52116-28	156
29	MS52116-29	168
30	MS52116-30	180
31	MS52116-31	192
32	MS52116-32	204
33	MS52116-33	216
34	MS52116-34	228
35	MS52116-35	240
36	MS52116-36	252
37	MS52116-37	264
38	MS52116-38	276
39	MS52116-39	288
40	MS52116-40	300

FIGURE 3. Shaft assembly, flexible, plastic, tachometer and speedometer - optional construction - Continued.

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NOTES:
1. All dimensions are shown in inches.
2. Unless otherwise specified, all tolerances shall be ±.010.
3. Material: Washer, fiber, vulcanized; grade: commercial; form: sheet; in accordance with ASTM D710. Washer is optional if tip and ferrule design incorporates an alternate method of cable tip retention.
4. For associated shaft assemblies, see figure 1.
5. Outer layer of core wire shall be wound in the direction of right hand lay.

FIGURE 4. Core, flexible shaft assembly, tachometer and speedometer.

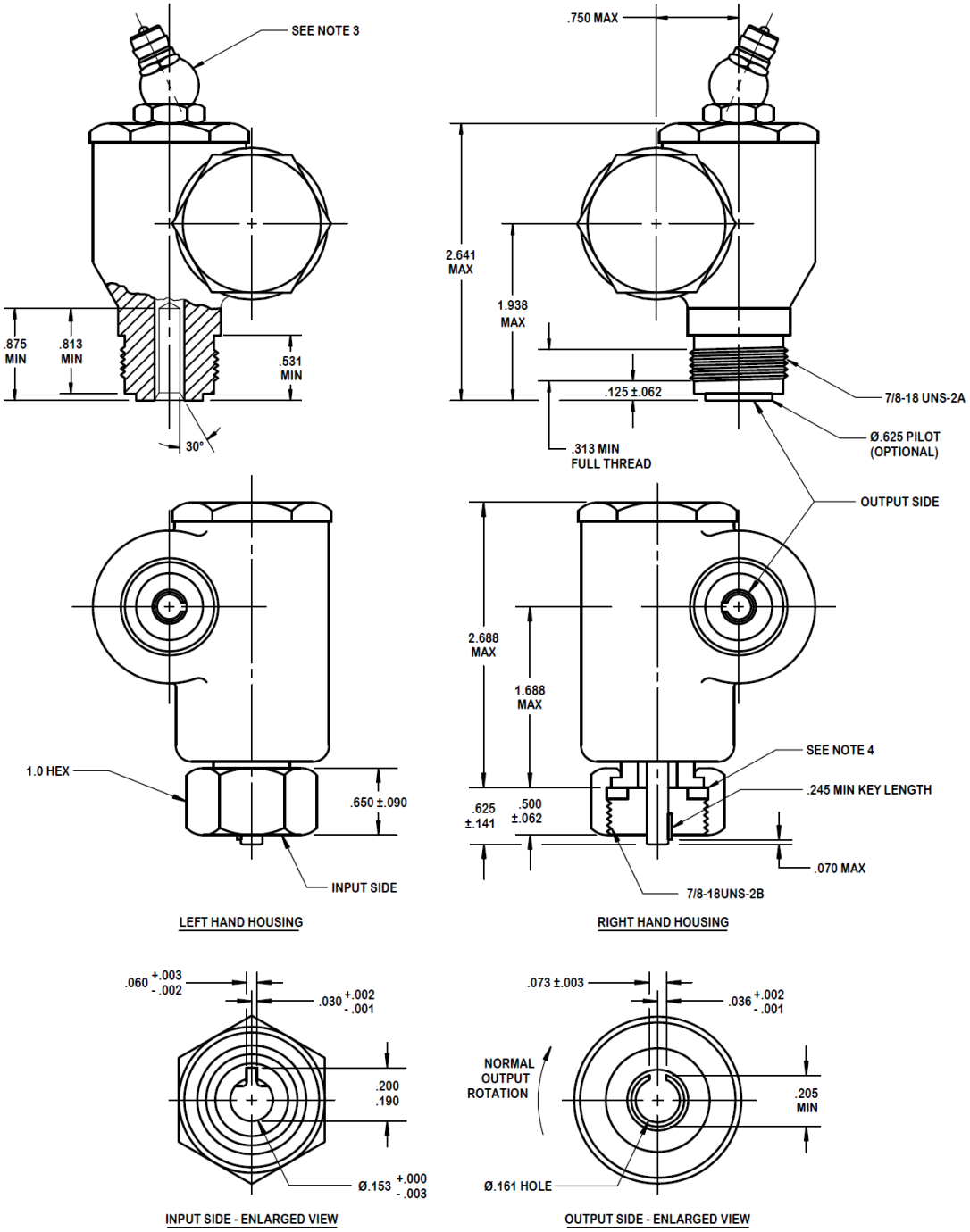
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Size no.	Former MS part no.	Former Army part no.	Core length
01	MS51072-1	8749985	18.7
02	MS51072-2	7976737	23.7
03	MS51072-3	7418497	41.2
04	MS51072-4	7527483	56.2
05	MS51072-5	7064607	63.7
07	MS51072-7	7064520	68.7
08	MS51072-8	7418498	73.7
09	MS51072-9	7418491	78.7
10	MS51072-10	7527481	86.2
11	MS51072-11	7415830	91.2
12	MS51072-12	7418490	96.2
13	MS51072-13	7527482	101.2
14	MS51072-14	7984852	108.7
15	MS51072-15	7984853	145.2
16	MS51072-16	7064524	157.2
17	MS51072-17	8366051	169.2
18	MS51072-18	8741685	217.2
19	MS51072-19	7355827	121.2
20	MS51072-20	8345796	139.2
21	MS51072-21	-	14.2
22	-	7416396	51.2
23	-	7355830	58.7
24	-	7355828	181.2

NOTE: For associated shaft assemblies see figure 1. Outer layer of core wire shall be wound in the direction of right hand lay.

FIGURE 4. Core, flexible shaft assembly, tachometer and speedometer - Continued.

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NOTES:

1. All dimensions are shown in inches.
2. Unless otherwise specified, all tolerances shall be ±.010, ±1° on angular dimensions..
3. Lubricate with grease conforming to MIL-PRF-23827. Sealed lubrication fittings are optional.
4. For gasket material and dimensions, see former Army part number 7539688.
5. The body shape shown is provided as a general example only and does not imply the exclusion of alternative body shapes.

FIGURE 5. Adapter, speedometer and tachometer, 90-degree drive.

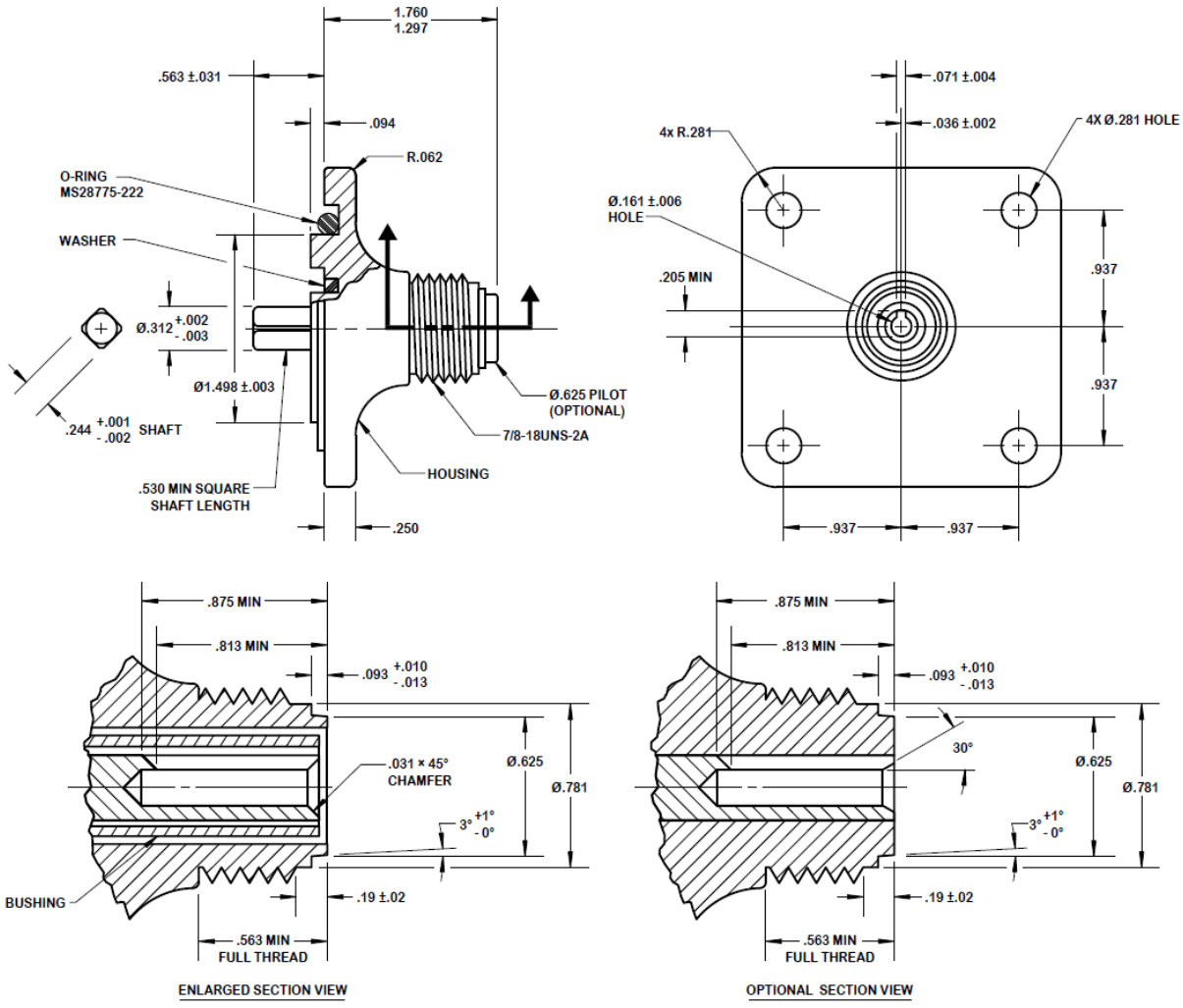
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Size no.	Former MS part no.	Former Army part no.	RPM ratio output/ input	Input shaft rotation when output shaft is rotated clockwise[1,2]	Housing type
01	MS39130-1	-	0.158	Clockwise	Right hand
02	MS39130-2	-	0.166	Counter-clockwise	Right hand
03	MS39130-3	7372726	0.200	Clockwise	Right hand
04	MS39130-4	-	0.200	Counter-clockwise	Right hand
05	MS39130-5	10900477	0.210	Clockwise	Left hand
06	MS39130-6	10917230	0.222	Clockwise	Right hand
07	MS39130-7	-	0.250	Clockwise	Right hand
08	MS39130-8	-	0.250	Counter-clockwise	Right hand
09	MS39130-9	-	0.300	Clockwise	Right hand
10	MS39130-10	-	0.300	Counter-clockwise	Right hand
11	MS39130-11	10900443	0.364	Clockwise	Right hand
12	MS39130-12	10921625	0.364	Counter-clockwise	Right hand
13	MS39130-13	-	0.416	Clockwise	Right hand
14	MS39130-14	-	0.421	Counter-clockwise	Right hand
15	MS39130-15	8340483	0.500	Clockwise	Right hand
16	MS39130-16	-	0.500	Counter-clockwise	Right hand
17	MS39130-17	6228369	0.500	Clockwise	Left hand
18	MS39130-18	6228367	0.500	Counter-clockwise	Left hand
19	MS39130-19	-	0.611	Clockwise	Right hand
20	MS39130-20	-	0.611	Counter-clockwise	Right hand
21	MS39130-21	-	0.705	Clockwise	Right hand
22	MS39130-22	-	0.714	Counter-clockwise	Right hand
23	MS39130-23	-	0.833	Counter-clockwise	Right hand
24	MS39130-24	-	0.909	Clockwise	Right hand
25	MS39130-25	7983058	1.000	Clockwise	Right hand
26	MS39130-26	-	1.000	Counter-clockwise	Right hand
27	MS39130-27	7987721	1.000	Clockwise	Left hand
28	MS39130-28	6209787	1.000	Counter-clockwise	Left hand
29	MS39130-29	-	1.100	Clockwise	Right hand
30	MS39130-30	-	1.100	Counter-clockwise	Right hand

[1]Shaft rotation is determined when looking into the input or output connector.
[2]Previous revisions of this table referred to the input shaft rotation with respect to output side as "same" (counter-clockwise) and "reverse" (clockwise).

FIGURE 5. Adapter, speedometer and tachometer, 90-degree drive - Continued.

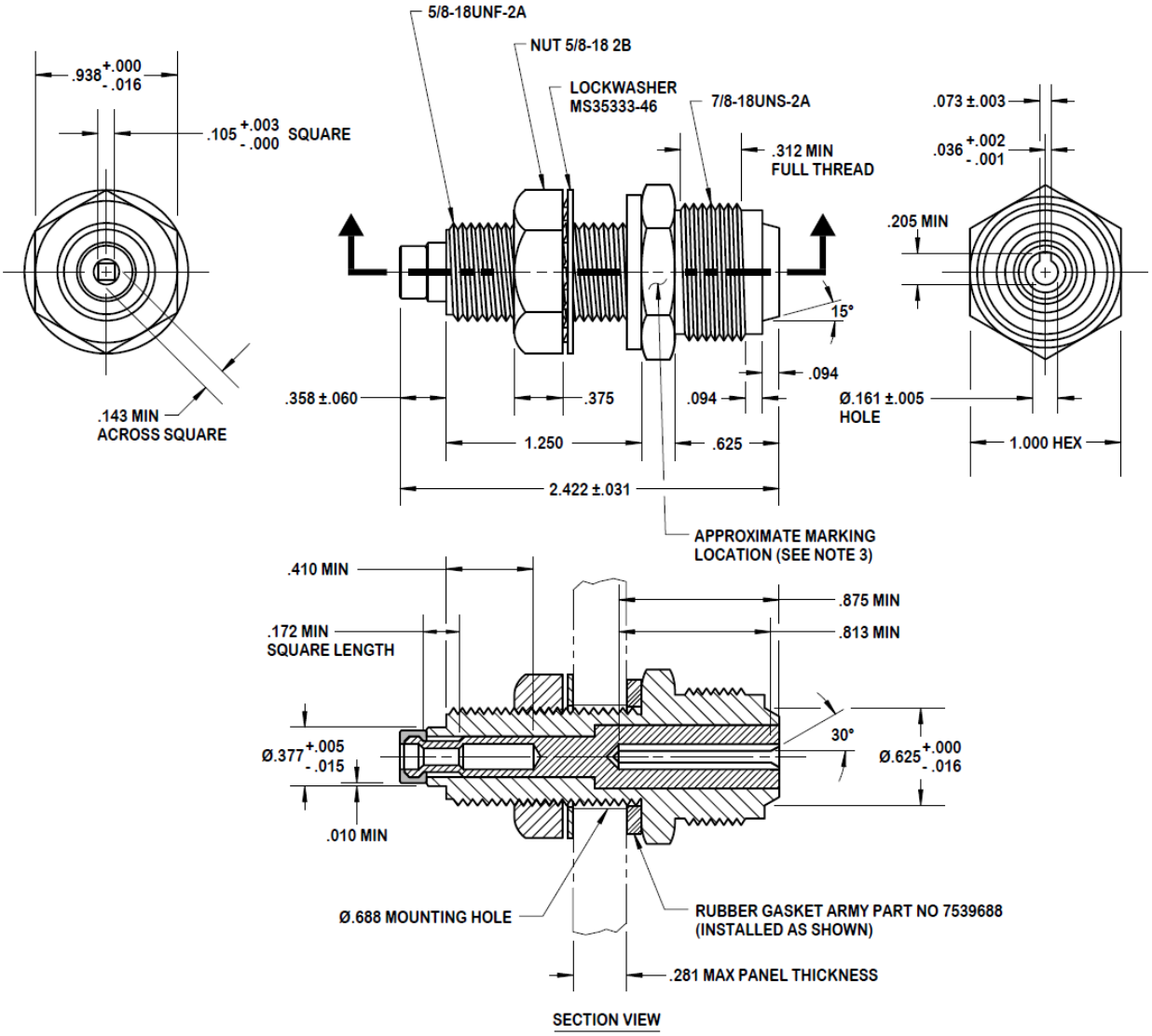
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NOTES:
1. All dimensions are shown in inches.
2. Unless otherwise specified, all tolerances shall be ±.010, ±2° on angular dimensions.

FIGURE 6. Adapter, speedometer and tachometer, straight drive.

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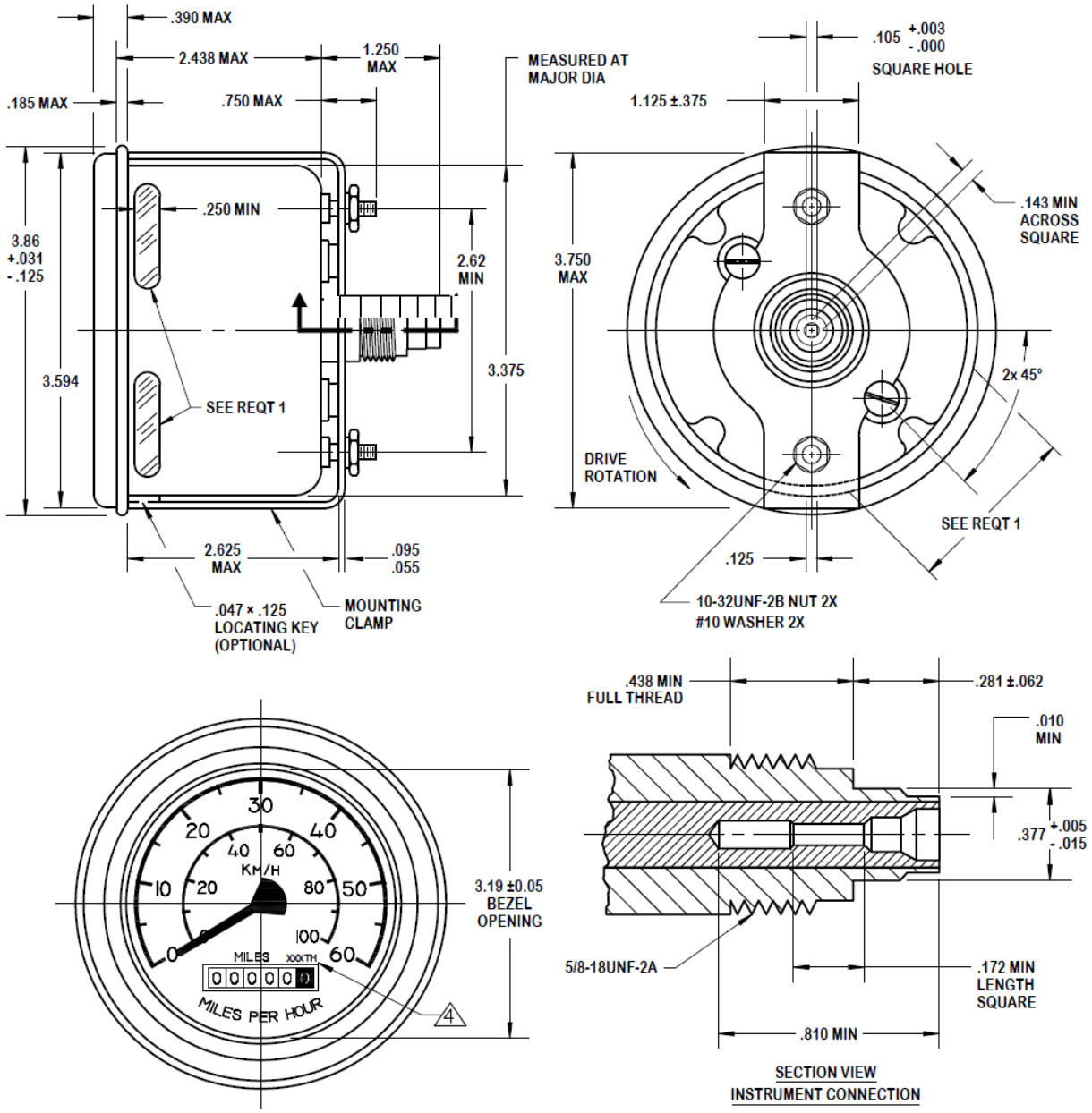
SECTION A-A

NOTES:
1. All dimensions are shown in inches.
2. Unless otherwise specified, all tolerances shall be ±.0106, ±1° on angular dimensions.
3. Marking shall consist of MS part number and the manufacturer's identification in accordance with MIL-STD-130.

Size no.	Former MS part no.	Former Army part no.	Description	Application
01	MS53099-1	7983056	Without rubber gasket	Non-waterproof
02	MS53099-2	-	With rubber gasket	Waterproof

FIGURE 7. Adapter, coupling/bulkhead, tachometer and speedometer drive.

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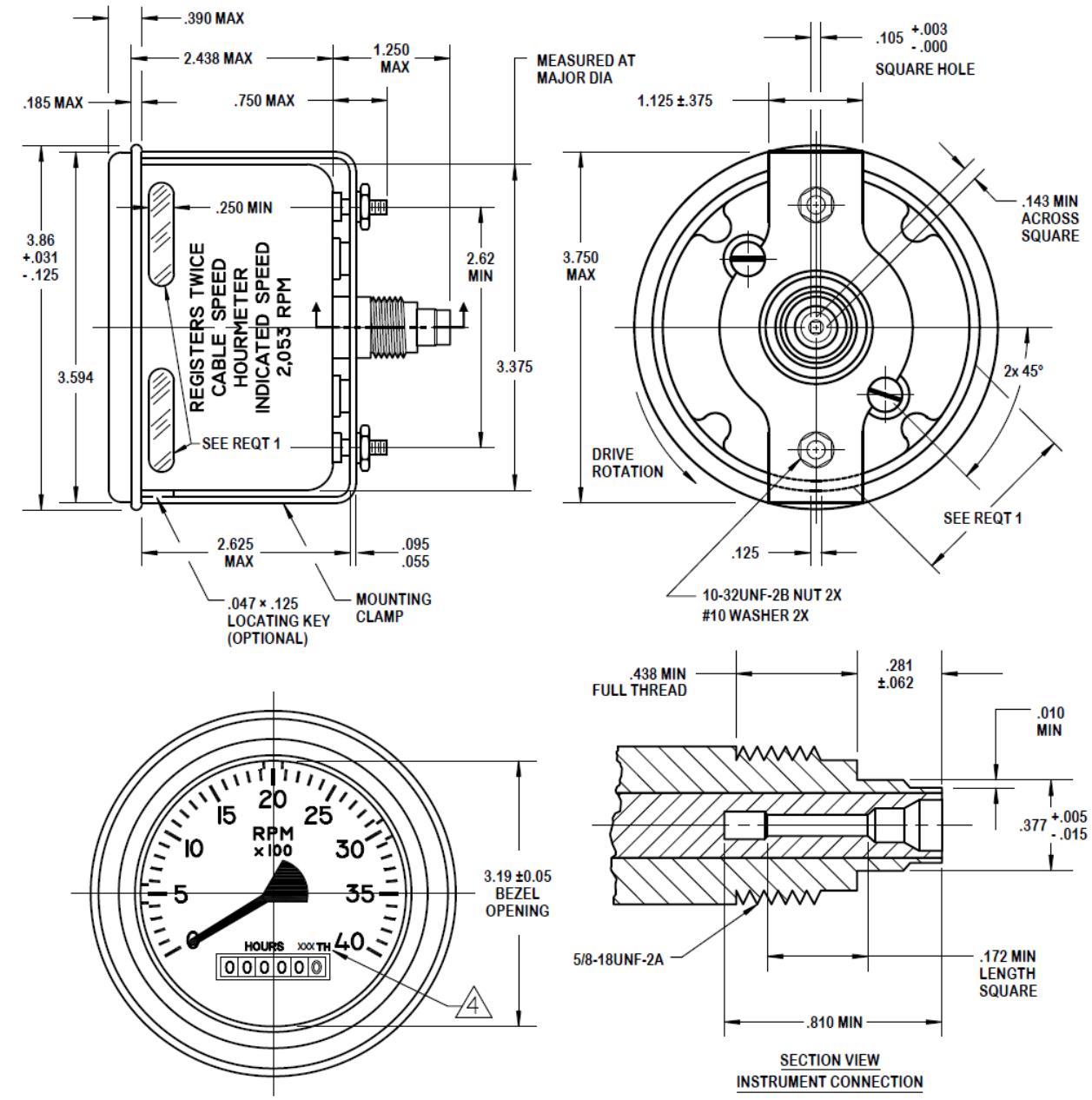


NOTES:

1. All dimensions are shown in inches.
2. Unless otherwise specified, all tolerances shall be ±.015, ±1° on angular dimensions.
3. Four illuminating windows shall be 1.86 minimum chordal length, equally spaced 45º of off horizontal centerline. Optional construction may include 360º illuminating window or fully transparent casing.
4. The specification requires either 1/100th or 1/10th of an hour increments.

FIGURE 8. Speedometer (0-60 mph).

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NOTES:

1. All dimensions are shown in inches.
2. Unless otherwise specified, all tolerances shall be ±.015, ±1° on angular dimensions.
3. Four illuminating windows shall be 1.86 minimum chordal length, equally spaced 45º of off horizontal centerline. Optional construction may include 360º illuminating window or fully transparent casing.

4. The specification requires either 1/100th or 1/10th of an hour increments.

FIGURE 9. Tachometer, fixed mounting (4,000 rpm).

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Custodians:
Army - AT
Navy - MC
DLA - GS

Preparing Activity:
DLA – GS3

(Project 6680-2024-003)

Review Activity:
Army - MI

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